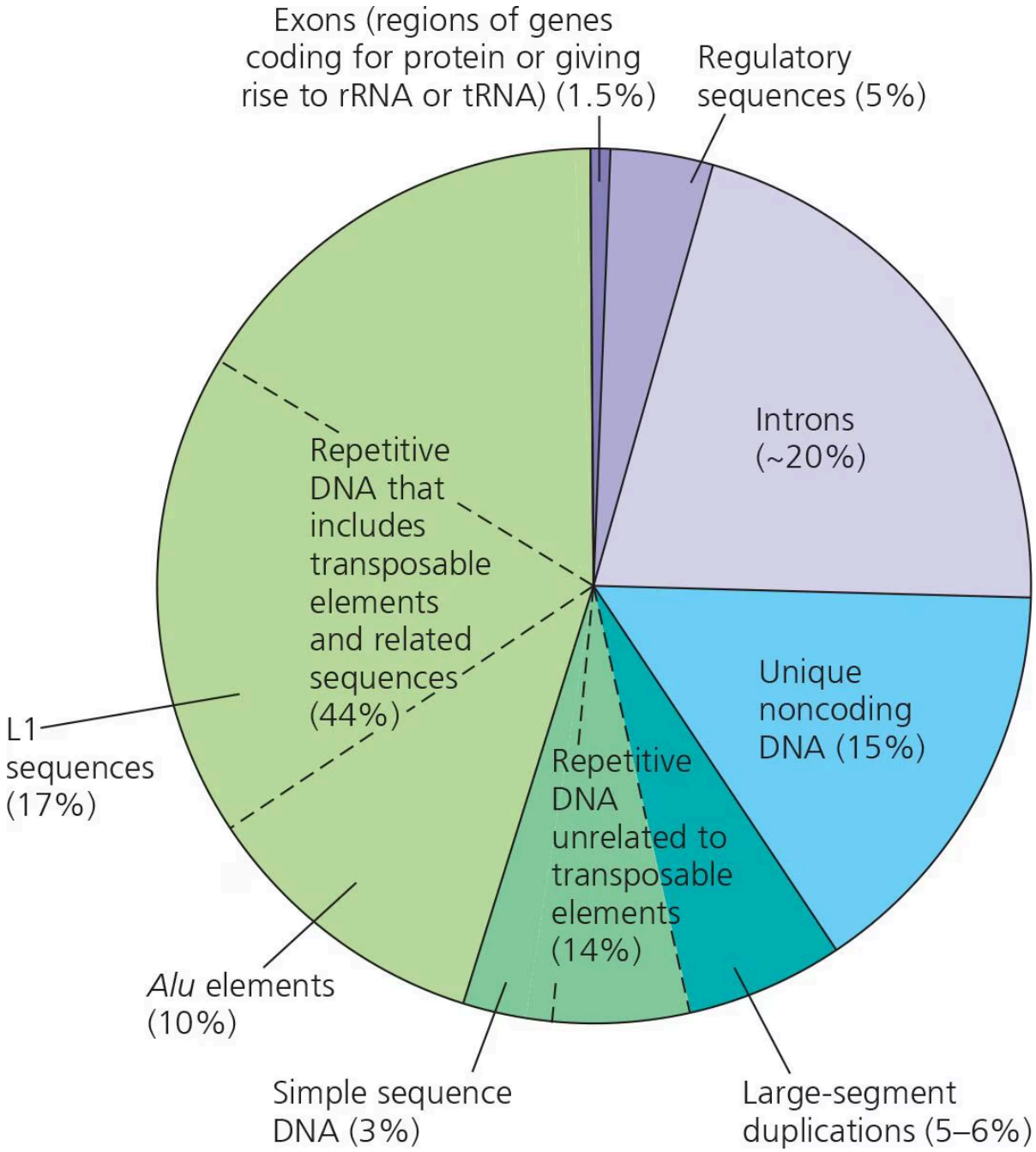
21.4: Multicellular eukaryotes have a lot of noncoding DNA and many multigene families

We have spent most of our time focusing on genes that code for proteins. Yet the coding regions of these genes and the genes for small RNAs like tRNAs make up a small portion of most multicellular eukaryotic genomes. For example, only a tiny part—about 1.5%—codes for proteins or is transcribed into rRNAs or tRNAs. **Figure 21.6** shows what is known about the makeup of the remaining 98.5% of the genome.

Figure 21.6

Types of DNA sequences in the human genome.



Pie chart of types of D N A sequences in the human genome.

Light green wedge: 44% of the pie chart is repetitive D N A that includes transposable elements and related sequences. Of this 44%, 17% is labeled L1 sequences and 10% is labeled Alu elements.

Green wedge: 14% is repetitive D N A unrelated to transposable elements, of which 3% is simple sequence D N A.

Dark green wedge: 5-6% is large-segment duplications. Blue wedge: 15% is unique non-coding D N A.

Lavender wedge: 20% is introns.

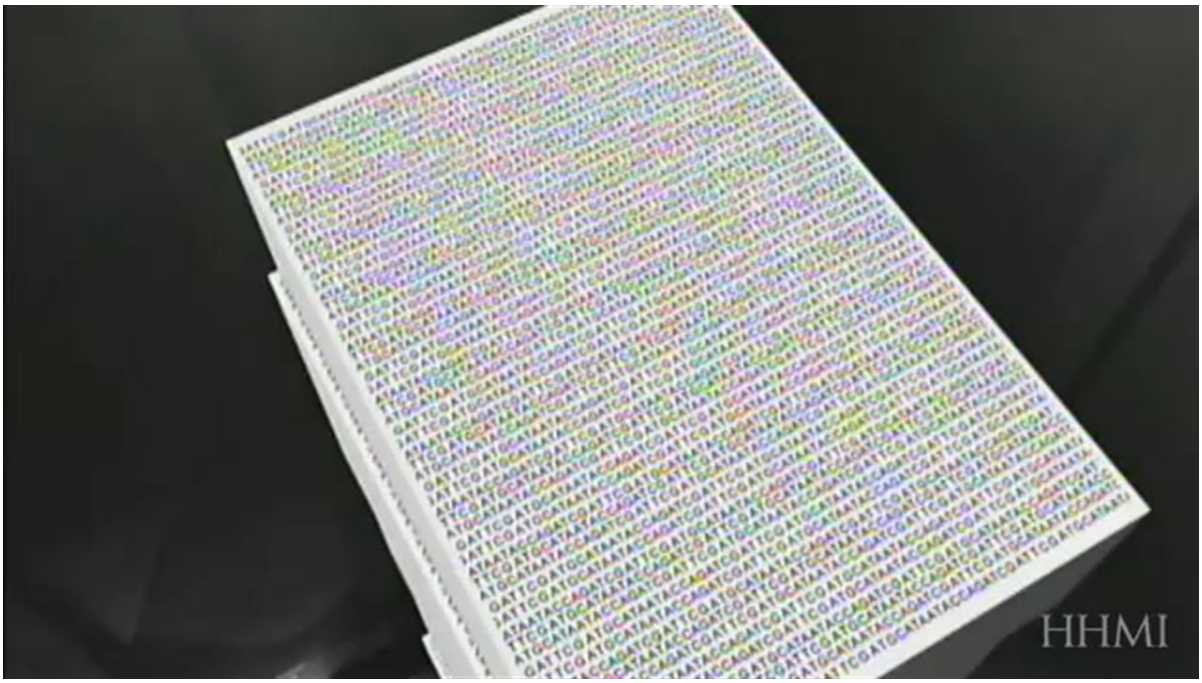
Purple wedge: 5% is regulatory sequences.

Dark purple wedge: 1.5% is Exons (regions of genes coding for proteins or giving rise to rR N A or t R N A).

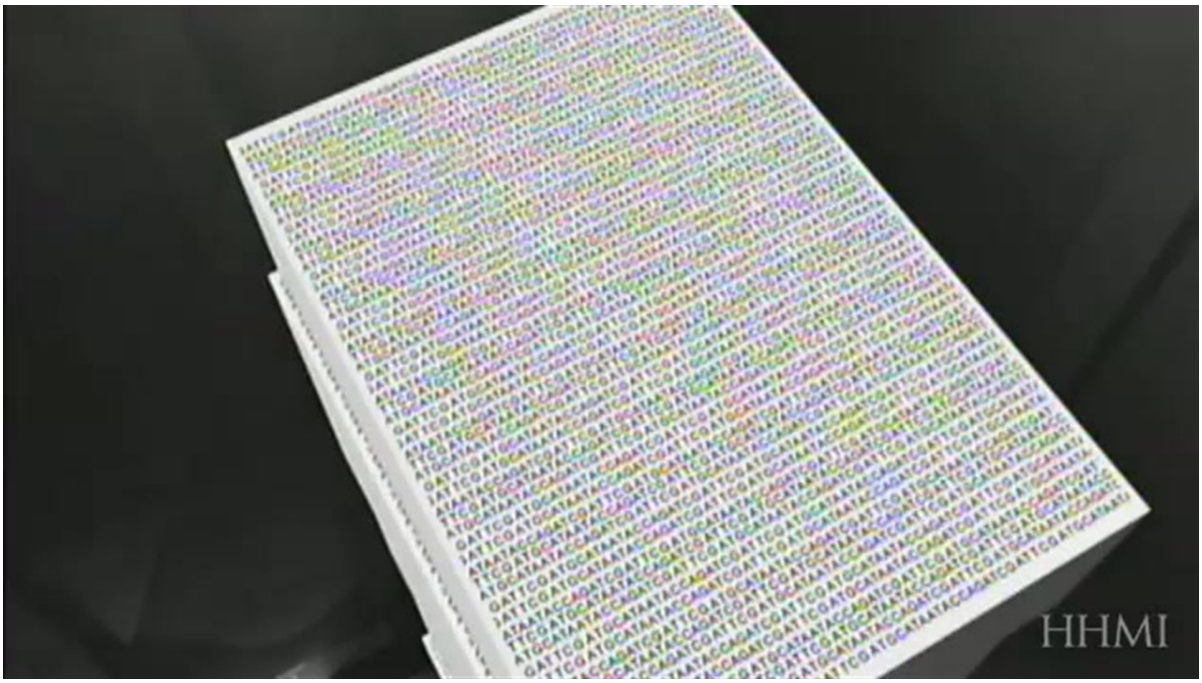
The gene sequences that code for proteins or are transcribed into rRNA or tRNA molecules make up only about 1.5% of the human genome (dark purple in the pie chart), while introns and regulatory sequences associated with genes (lighter purple) make up about a quarter. The vast majority of the human genome does not code for proteins (although much of it gives rise to RNAs), and a large amount is repetitive DNA (dark and light green and teal).

HHMI BioInteractive Animation

Coding Sequences in DNA



Play



Play

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Gene-related regulatory sequences and introns account, respectively, for 5% and about 20% of the human genome. The rest, located between functional genes, includes some unique (single-copy) noncoding DNA, such as gene fragments and **pseudogenes**, former genes that have accumulated mutations over a long time and no longer produce functional proteins. (The genes that produce small noncoding RNAs are a tiny percentage of the genome, distributed between the 20% introns and the 15% unique noncoding DNA.) Most of the DNA between functional genes, however, is **repetitive DNA**, which consists of sequences that are present in multiple copies in the genome.

The bulk of many eukaryotic genomes consists of DNA sequences that neither code for proteins nor are transcribed to produce RNAs with known functions; this noncoding DNA was often described in the past as "junk DNA." However, genome comparisons over the past ten years have revealed the persistence of this DNA in diverse genomes over many hundreds of generations. For example, the genomes of humans, rats, and mice contain almost 500 regions of noncoding DNA that are *identical* in sequence in all three species. This is a higher level of sequence conservation than is seen for protein-coding regions in these species, strongly suggesting that the noncoding regions have important functions. The results of the ENCODE project discussed earlier have underscored the key roles played by much of this noncoding DNA. Next, we'll examine how genes and noncoding DNA sequences are organized within genomes of multicellular eukaryotes, using the human genome as our main example. Genome organization tells us a lot about how genomes have evolved and continue to evolve, as we'll see in Concept 21.5.

Transposable Elements and Related Sequences

Both prokaryotes and eukaryotes have stretches of DNA that can move from one location to another within the genome. These stretches are known as *transposable genetic elements*, or simply **transposable elements**. During the process called *transposition*, a transposable element moves from one site in a cell's DNA to a different target site by a type of recombination process. Transposable elements are sometimes called "jumping genes," but actually they never entirely detach from the cell's DNA. Instead, the original and new DNA sites are brought very close together by enzymes and other proteins that bend the DNA. Surprisingly, about 75% of human repetitive DNA (44% of the entire human genome) is made up of transposable elements and sequences related to them.

The first evidence for wandering DNA segments came from American geneticist Barbara McClintock's breeding experiments with calico corn (maize) in the 1940s and 1950s **(Figure 21.7)**. Tracking corn plants through many generations, McClintock analyzed changes in the color of corn kernels. The patterns she saw led her to propose that there were genetic elements capable of moving from other locations in the genome into the genes for kernel color, disrupting the genes and changing the kernel color. McClintock's hypothesis provoked great interest among her maize colleagues, but most other scientists thought the phenomenon she had observed might occur only in maize. Her careful work and insightful ideas were finally validated many years later when transposable elements were found in bacteria. In 1983, at the age of 81, McClintock received the Nobel Prize for her pioneering research.

Figure 21.7

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The effect of transposable elements on corn kernel color.
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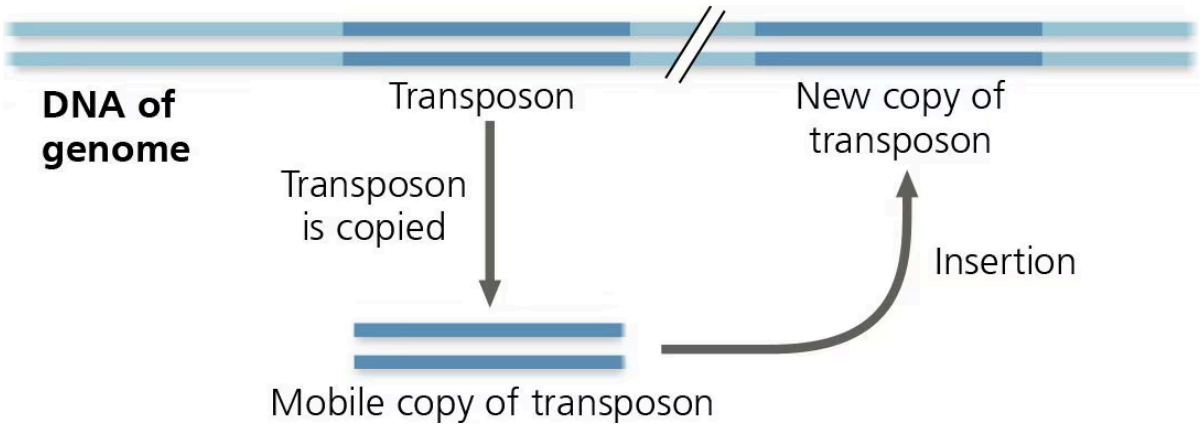
Barbara McClintock first proposed the idea of mobile genetic elements after observing variegations in the color of the kernels on a corn cob (top right).

Movement of Transposons and Retrotransposons

Eukaryotic transposable elements are of two types. The first type, **transposons**, move within a genome by means of a DNA intermediate. Transposons can move by a “cut-and-paste” mechanism, which removes the element from the original site, or by a “copy-and-paste” mechanism, which leaves a copy behind **(Figure 21.8)**. Both mechanisms require an enzyme called *transposase*, which is generally encoded by the transposon.

Figure 21.8

Transposon movement.



Movement of transposons by either the copy-and-paste mechanism (shown here) or the cut-and-paste mechanism involves a double-stranded DNA intermediate that is inserted into the genome.

VISUAL SKILLS *How would this figure differ if it showed the cut-and-paste mechanism?*

*For suggested answer, see*Appendix A.**

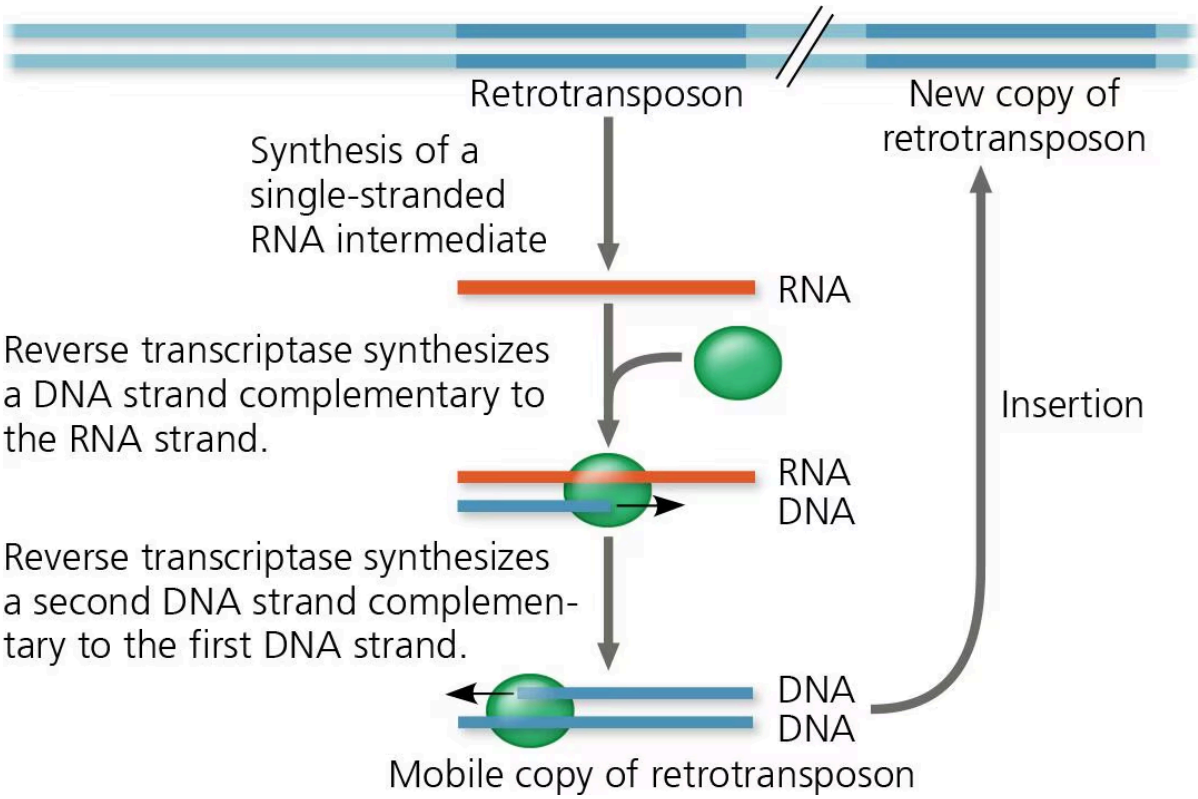
Interview

Interview with Virginia Walbot: Plant genetics and development

Most transposable elements in eukaryotic genomes are of the second type, **retrotransposons**, which move by means of an RNA intermediate that is a transcript of the retrotransposon DNA. Thus, retrotransposons always leave a copy at the original site during transposition **(Figure 21.9)**. To insert at another site, the RNA intermediate is first converted back to DNA by reverse transcriptase, an enzyme encoded by the retrotransposon. (Reverse transcriptase is also encoded by retroviruses, as you learned in Concept 19.2. In fact, retroviruses may have evolved from retrotransposons, or vice versa.) Another cellular enzyme catalyzes insertion of the reverse-transcribed DNA at a new site.

Figure 21.9

Retrotransposon movement.



Double-strand D N A with retrotransposon region labeled (both strands). Single-stranded R N A intermediate is synthesized from retrotransposon. Reverse transcriptase synthesizes a D N A strand complementary to the R N A strand. Reverse transcriptase synthesizes a second D N A strand complementary to the first D N A strand. A mobile copy of the retrotransposon is inserted into the D N A downstream of the original retrotransposon.

Movement begins with synthesis of a single-stranded RNA intermediate. The remaining steps are essentially identical to part of the retrovirus replicative cycle (see Figure 19.8).

Sequences Related to Transposable Elements

Multiple copies of transposable elements and sequences related to them are scattered throughout eukaryotic genomes. A single unit is usually hundreds to thousands of base pairs long, and the dispersed copies are similar but usually not identical to each other. Some of these are transposable elements that can move; the enzymes required for this movement may be encoded by any transposable element, including the one that is moving. Others are related sequences that have lost the ability to move altogether. Transposable elements and related sequences make up 25–50% of most mammalian genomes (see Figure 21.6) and even higher percentages in amphibians and many plants. In fact, the very large size of some plant genomes is accounted for by extra transposable elements rather than by extra genes. For example, transposable elements make up 85% of the corn genome!

In humans and other primates, a large portion of transposable element–related DNA consists of a family of similar sequences called *Alu elements*. These sequences alone account for approximately 10% of the human genome. *Alu* elements are about 300 nucleotides long, much shorter than most functional transposable elements, and they do not code for any protein. However, many *Alu* elements are transcribed into RNA, and at least some of these RNAs are thought to help regulate gene expression.

An even larger percentage (17%) of the human genome is made up of a type of retrotransposon called LINE-1, or L1. These sequences are much longer than *Alu* elements—about 6,500 base pairs—and typically have a very low rate of transposition. However, researchers working with mice have discovered that transcription of L1 retrotransposons is crucial for the development of early (one- and two-cell stage) embryos. They have proposed that transcription of L1 retrotransposons may affect the chromatin structure in ways important for embryonic development.

Although some transposable elements encode proteins, these proteins do not carry out normal cellular functions. Therefore, transposable elements are usually included in the "noncoding" DNA category, along with other repetitive sequences.

Other Repetitive DNA, Including Simple Sequence DNA

Repetitive DNA that is not related to transposable elements has probably arisen from mistakes during DNA replication or recombination. Such DNA accounts for about 14% of the human genome (see Figure 21.6). About a third of this (5– 6% of the human genome) consists of duplications of long stretches of DNA, with each unit ranging from 10,000 to 300,000 base pairs. These long segments seem to have been copied from one chromosomal location to another site on the same or a different chromosome and probably include some functional genes.

In contrast to scattered copies of long sequences, stretches of DNA known as **simple sequence DNA** contain many copies of tandemly repeated short sequences, as in the following example (showing one DNA strand only):

...GTTACGTTACGTTACGTTACGTTACGTTAC...

In this case, the repeated unit (GTTAC) consists of 5 nucleotides, but the number can range from 2 to 500. When the unit contains 2–5 nucleotides, the series of repeats is called a ****__short tandem repeat__, or **STR**; we discussed the use of STR analysis in preparing genetic profiles by use of PCR in Concept 20.4 (see Figure 20.24). The number of copies of the repeated unit can vary from site to site within a given genome. There could be as many as several hundred thousand repetitions of the GTTAC unit at one site, but only half that number at another. STR analysis is performed on sites selected because they have relatively few repeats.

The repeat number varies from person to person, and since humans are diploid, each person has two alleles per repeat site; these can differ in repeat number. This diversity produces the variation represented in the genetic profiles that result from STR analysis.

Simple sequence DNA makes up 3% of the human genome, much of it located at chromosomal telomeres and centromeres, where it may play a structural role. The DNA at centromeres is essential for the separation of chromatids in cell division (see Concept 12.2) and, along with simple sequence DNA located elsewhere, may also help organize the chromatin within the interphase nucleus. The simple sequence DNA located at telomeres binds proteins that protect chromosomal ends from degradation and from joining to other chromosomes.

Short repetitive sequences like those described here provide a challenge for whole-genome shotgun sequencing because the presence of many short repeats hinders accurate reassembly of fragment sequences by computers. Regions of simple sequence DNA account for much of the uncertainty present in estimates of whole-genome sizes and are the reason some sequences are considered “permanent drafts.”

Genes and Multigene Families

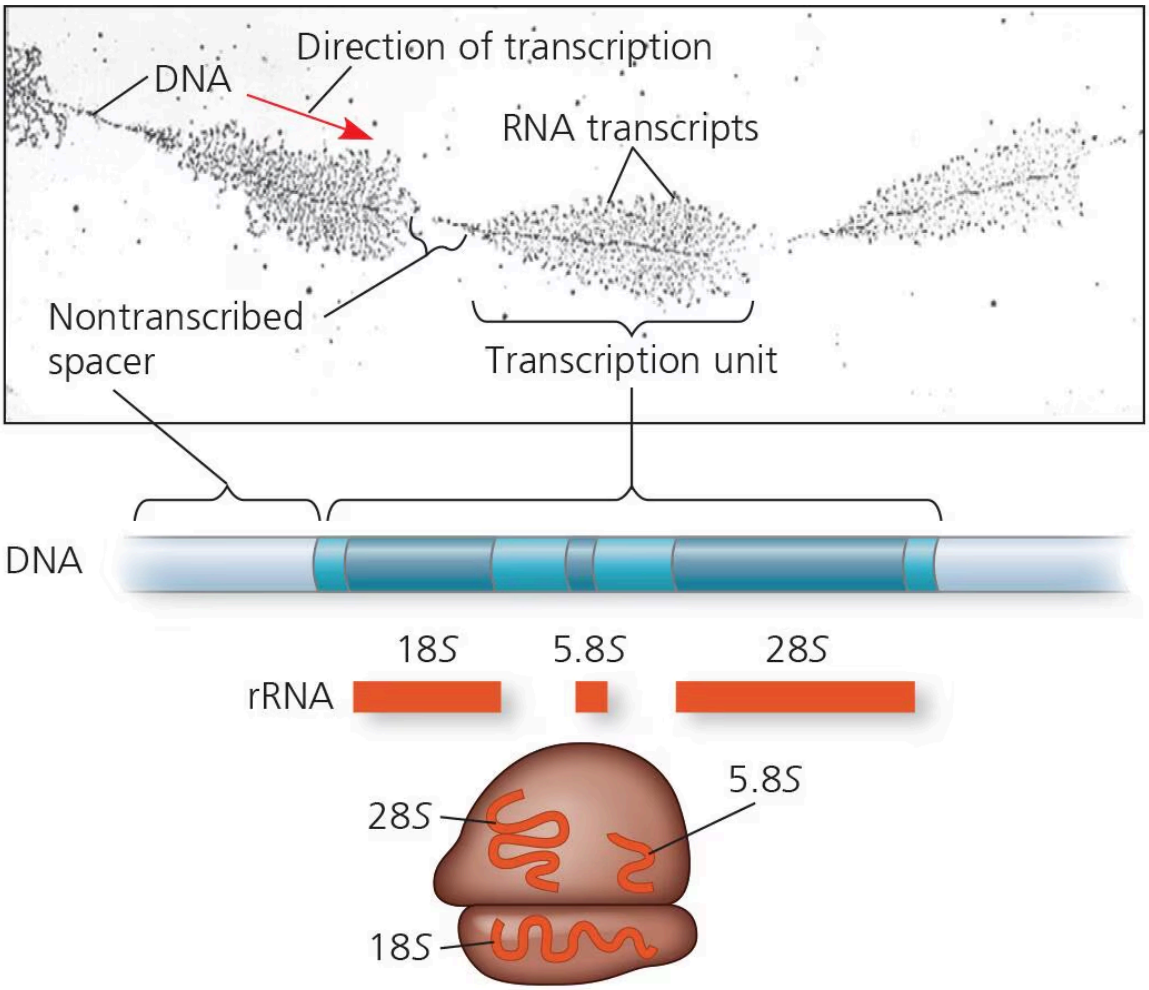
Now, let’s take a look at genes. Recall that DNA sequences that code for proteins or give rise to tRNA or rRNA make up only 1.5% of the human genome (see Figure 21.6). If we include introns and regulatory sequences, the total amount of DNA that is gene-related—coding and noncoding—constitutes about 25% of the human genome. Put another way, only about 6% (1.5% out of 25%) of the length of the average gene is represented in the final gene product.

Many eukaryotic genes are present as unique sequences, with only one copy per haploid set of chromosomes. But in the human genome and the genomes of many other animals and plants, these unique genes make up less than half of the total gene-related DNA. The rest occur in **multigene families**, collections of two or more identical or very similar genes.

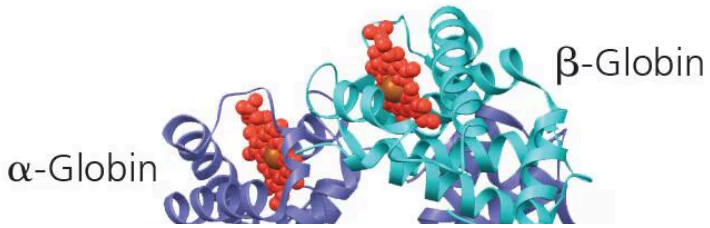
In multigene families that consist of *identical* DNA sequences, those sequences are usually clustered tandemly and, with the notable exception of the genes for histone proteins, have RNAs as their final products. An example is the family of identical DNA sequences that each include the genes for the three largest rRNA molecules **(Figure 21.10a)**. These rRNA molecules are transcribed from a single transcription unit that is repeated tandemly hundreds to thousands of times in one or several clusters in the genome of a multicellular eukaryote. The many copies of this rRNA transcription unit help cells to quickly make the millions of ribosomes needed for active protein synthesis. The primary transcript is cleaved to yield three rRNA molecules, which combine with proteins and one other kind of rRNA (5*S* rRNA) to form ribosomal subunits.

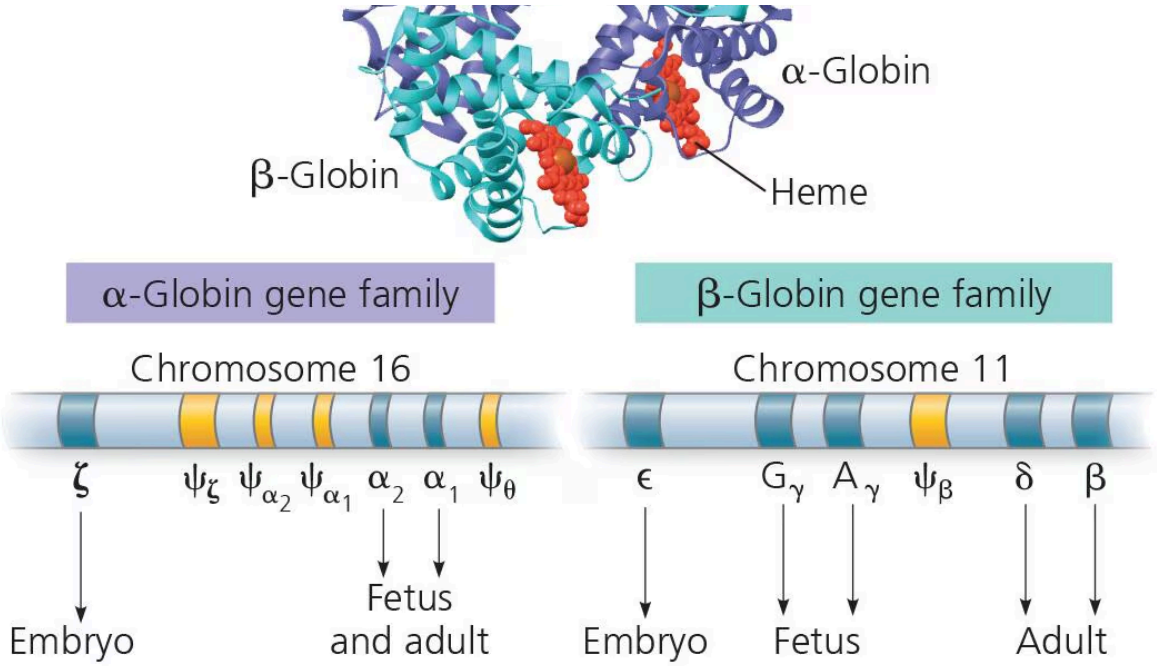
Figure 21.10

Gene families.



(a) Part of the ribosomal RNA gene family. The TEM at the top shows three of the hundreds of copies of rRNA transcription units in the rRNA gene family of a salamander genome. Each "feather" corresponds to a single unit being transcribed by about 100 molecules of RNA polymerase (dark dots along the DNA), moving left to right (red arrow). The growing RNA transcripts extend from the DNA, accounting for the feather-like appearance. In the diagram of a transcription unit below the TEM, the genes for three types of rRNA (darker blue) are adjacent to regions that are transcribed but later removed (medium blue). A single transcript is processed to yield one of each of the three rRNAs (red), key components of the ribosome.





(b) The human α-globin and β-globin gene families. Adult hemoglobin is composed of two α-globin and two β-globin polypeptide subunits, as shown in the molecular model. The genes (darker blue) encoding α- and β-globins are found in two families, organized as shown here. The noncoding DNA (light blue) separating the functional genes within each family includes pseudogenes (ψ; gold), versions of the functional genes that no longer encode functional polypeptides. Genes and pseudogenes are named with Greek letters, as you have seen previously for the α- and β-globins. Some genes are expressed only in the embryo or fetus.

Parts are as follows.

Part a. Part of the ribosomal R N A gene family. The TEM at the top shows three of the hundreds of copies of rR N A transcription units in the rR N A gene family of a salamander genome. Each feather corresponds to a single unit being transcribed by about 100 molecules of R N A polymerase (dark dots along the D N A), moving left to right (red arrow). The growing R N A transcripts extend from the D N A, accounting for the feather-like appearance. In the diagram of a transcription unit below the TEM, the genes for three types of rR N A (darker blue) are adjacent to regions that are transcribed but later removed (medium blue). A single transcript is processed to yield one of each of the three rR N As (red), key components of the ribosome.

Part b. The human alpha-globin and beta-globin gene families. Adult hemoglobin is composed of two alpha-globin and two beta-globin polypeptide subunits, as shown in the molecular model. The genes (darker blue) encoding alpha- and beta-globins are found in two families, organized as shown here. The noncoding D N A

(light blue) separating the functional genes within each family includes pseudogenes (Greek letter psi; gold), versions of the functional genes that no longer encode functional polypeptides. Genes and pseudogenes are named with Greek letters, as you have seen previously for the alpha- and beta-globins. Some genes are expressed only in the embryo or fetus. Ribbon model of hemoglobin molecule with four heme groups, two alpha-globin subunits and 2 beta-globin subunits. Chromosomes 16 and 11 also diagramed.

Key: Light blue = noncoding regions. Dark blue = gene. Gold = pseudogene. Alpha-globin gene family. Chromosome 16 regions, left to right: Noncoding region; zeta gene with arrow to embryo; noncoding region; psi zeta gene; noncoding region; psi alpha 2 pseudogene; noncoding region; psi alpha 1 pseudogene; noncoding region; alpha 2 gene with arrow to fetus and adult; noncoding region; alpha 1 gene with arrow to fetus and adult; noncoding region; psi theta pseudogene; noncoding region. Beta-globin gene family. Chromosome 11 regions, left to right: Noncoding region; epsilon gene with arrow to embryo; noncoding region; gamma G gene with arrow to fetus; noncoding region; gamma A gene with arrow to fetus; noncoding region; psi beta pseudogene; noncoding region; delta gene with arrow to adult; noncoding region; beta gene with arrow to adult; noncoding region.

VISUAL SKILLS *In the TEM at the top of part (a), how could you determine the direction of transcription if it weren't indicated by the red arrow?*

*For suggested answer, see*Appendix A*.*

The classic examples of multigene families of *nonidentical* genes are two related families of genes that encode globins, a group of proteins that include the α and β polypeptide subunits of hemoglobin. One family, located on chromosome 16 in humans, encodes various forms of α-globin; the other, on chromosome 11, encodes forms of β-globin **(Figure 21.10b)**. The different forms of each globin subunit are expressed at different times in development, allowing hemoglobin to function effectively in the changing environment of

the developing animal. In humans, for example, the embryonic and fetal forms of hemoglobin have a higher affinity for oxygen than the adult forms, ensuring the efficient transfer of oxygen from mother to fetus. Also found in the globin gene family clusters are several pseudogenes.

In Concept 21.5, we'll consider the evolution of these two globin gene families as we explore how arrangements of genes provide insight into the evolution of genomes. We'll also examine some processes that have shaped the genomes of different species over evolutionary time.